

Chapter 5f: Approximation of General Mapping

Fixed Point Theorems in Ordered Banach Spaces

Pathak - An Introduction to Nonlinear Analysis and Fixed Point Theory

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Definition 5.75-5.78: Fréchet Derivatives and Operators

Definition (Fréchet Differentiability)

A nonlinear operator $F : X \rightarrow X$ is Fréchet differentiable with respect to cone K at x_0 if:

$$F(x_0 + h) - F(x_0) = F'(x_0)h + w(x_0, h), \quad h \in K$$

where $\lim_{h \in K, \|h\| \rightarrow 0} \frac{\|w(x_0, h)\|}{\|h\|} = 0$.

Definition (Positive Operator)

A linear operator $F : X \rightarrow X$ is:

- 1 **Positive** if $F(K) \subseteq K$
- 2 **Strictly positive** if $F(\bar{K}) \subset K$ (interior)
- 3 **Strongly positive** if $F(\bar{K}) \subseteq \text{int}(K)$

Characteristic vectors are eigenvectors of F in cone K with nonnegative eigenvalues.

Theorem 5.145: Krasnoselski's Characteristic Vector Theorem

Theorem (Krasnoselski [345])

] Let T be a continuous linear positive operator mapping bounded subsets into compact sets.

If $T^p u \geq \alpha u$ for some nonzero u with $-u \notin K$, $u = v - w$ ($v, w \in K$), and some $p \in \mathbb{N}$, then T has at least one characteristic vector $x_0 \in K$ with:

$$Tx_0 = \eta_0 x_0$$

where $\eta_0 \geq p\sqrt{\alpha}$.

Key insight: Characteristic vectors correspond to positive eigenvectors of the operator T .

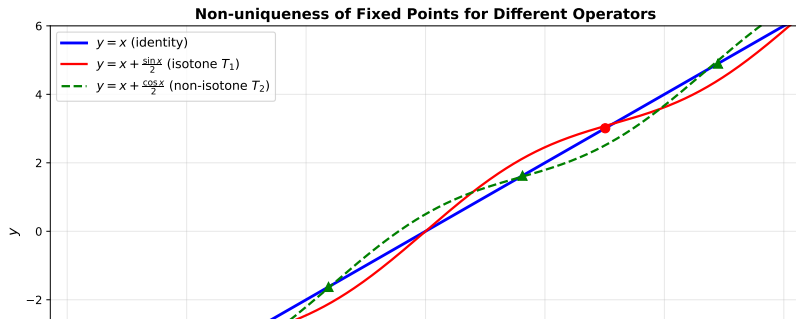
Example 5.50: Multiple Fixed Points

Consider $X = \mathbb{R}$ with $K = [0, \infty)$ and operator:

$$T_X = x + \frac{\sin x}{2}, \quad x \geq 0$$

This isotone operator has three fixed points:

$$x \in \left\{ 0, \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$



Definition 5.80-5.81: Minimal/Maximal and Majorant

Definition (Minimal/Maximal Fixed Points)

A fixed point x^ of $T : Y \rightarrow X$ is minimal if every fixed point y satisfies $x^* \leq y$ ($x^* \geq y$). There is at most one minimal (maximal) fixed point.*

Definition (Majorant/Minorant)

A mapping $\bar{F} : Y \rightarrow X$ is a majorant of F if $F(y) \leq \bar{F}(y)$ for all $y \in Y$. Minorants are defined by reversing the inequality.

Spectral radius: For linear operator $F : X \rightarrow X$,

$$r(F) = \lim_{k \rightarrow \infty} \|F^k\|^{1/k}$$

Theorem 5.150: Amann's Iterated Sequence

Theorem (Amann [17])

] Let (X, K) be an ordered Banach space with D order-convex. If $F : D \rightarrow X$ is increasing with $F(D)$ relatively compact, and there exist $\underline{y}, \bar{y} \in D$ with $\underline{y} \leq F(\underline{y})$ and $F(\bar{y}) \leq \bar{y}$, then F has a minimal fixed point $\underline{x} \in (\underline{y}, \bar{y}) \cap D$ with:

$$\underline{x} = \lim_{k \rightarrow \infty} F^k(\underline{y})$$

The sequence $\{F^k(\underline{y})\}$ is increasing.

Proof idea: Isotone property ensures monotone sequence convergence to fixed point.

Corollary 5.32-5.33: Minimal and Maximal Fixed Points

Corollary (Multiple Fixed Points)

If hypotheses of Theorem 5.150 hold, then:

- $\underline{x} = \lim_{k \rightarrow \infty} F^k(\underline{y})$ (*minimal fixed point*)
- $\bar{x} = \lim_{k \rightarrow \infty} F^k(\bar{y})$ (*maximal fixed point*)
- Sequences $\{F^k(\underline{y})\}$ and $\{F^k(\bar{y})\}$ are monotone

Key result: Isotone mappings produce monotone iteration sequences.

Theorems 5.151-5.153: Asymptotics and Nonisotone Operators

Theorem (Spectral Radius Condition)

If $F : K \rightarrow K$ is increasing, continuous, maps intervals to relatively compact sets, and has strongly positive Fréchet derivative $F'(0)$ with $r(F'(0)) > 1$, and continuous asymptotically linear majorant $\bar{F}$ with $r(F'(\infty)) < 1$, then F has at least one fixed point in $\text{int}(K)$.

Theorem (Nonisotone Fixed Points)

If F has Fréchet and asymptotic derivatives with appropriate spectral radius bounds, then nonzero fixed points exist under continuity/compactness conditions.

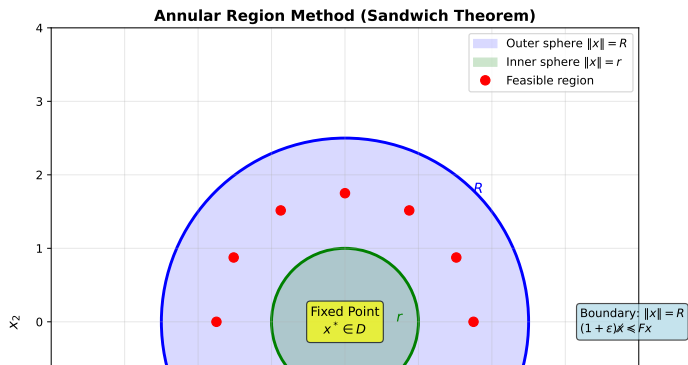
Generalizes fixed point theorems using asymptotic operator behavior.

Definition 5.82: Compression of a Cone

Definition (Compression)

A mapping $F : K \rightarrow K$ is a compression if $F(0) = 0$ and there exist $0 < r < R$ such that:

- ① $Fx \not\leq x$ for $x \in K$, $\|x\| \leq r$, $x \neq 0$ (expansion near 0)
- ② $(1 + \varepsilon)x \not\leq Fx$ for $x \in K$, $\|x\| \geq R$ (compression near ∞)



Theorem 5.157-5.158: Leggett-Williams Fixed Points

Theorem (Leggett-Williams [363])

] Let $F : K_R \rightarrow K$ be continuous with $F(0) = 0$, mapping bounded sets into relatively compact sets. If:

- ① For all $\varepsilon > 0$: $(1 + \varepsilon)x \not\leq Fx$ for $x \in K$, $\|x\| = R$*
- ② There exists null sequence $\{u_n\}$ in K_R such that if $x_n \rightarrow 0$ with $x_n \geq u_n$, there exists subsequence with $Fx_{n_k} \not\leq x_{n_k}$*

then F has a nonzero fixed point in K_R .

Extension: Multiple fixed point theorems (Williams-Leggett [607]).

Definitions 5.83-5.89: Partially Ordered Space Operators

Definition

- **Isotone:** $T(E) \rightarrow E$ preserves order: $x \preceq y \Rightarrow Tx \preceq Ty$
- **Compact:** $T(E)$ is relatively compact
- **Partially Continuous:** $\|Tx - Ty\| < \varepsilon$ when $\|x - y\| < \delta$ for comparable x, y
- **Partially Compact:** $T(C)$ relatively compact for all chains C
- **D-Lipschitzian:** $\|Tx - Ty\| \leq \varphi(\|x - y\|)$ for comparable elements

Compatibility condition: Order relation $\preceq$ and metric d compatible if nondecreasing sequences converging in metric imply pointwise convergence.

Theorem 5.168: Dhage's Hybrid Fixed Point Theorem

Theorem (Dhage [177])

] Let $(E, \preceq, \|\cdot\|)$ be a regular partially ordered complete normed linear space.

Let $\mathcal{P}, \mathcal{Q} : E \rightarrow E$ be nondecreasing operators where:

- ① $\mathcal{P}$ is partially bounded and partially nonlinear D -contraction*
- ② $\mathcal{Q}$ is partially continuous and partially compact*
- ③ There exists $x_0 \in E$ with $x_0 \preceq \mathcal{P}x_0 + \mathcal{Q}x_0$*

Then $\mathcal{P}x + \mathcal{Q}x = x$ has solution $x^ \in E$ with sequence $\{x_n\}_{n=0}^{\infty}$ (where $x_{n+1} = \mathcal{P}x_n + \mathcal{Q}x_n$) converging monotonically to x^* .*

Weakly Contractive Mappings: Definition

Definition (Weak Contraction)

A self-map $T : X \rightarrow X$ is weakly contractive if for all $x, y \in X$:

$$\|Tx - Ty\| \leq \|x - y\| - \psi(\|x - y\|)$$

where $\psi : [0, \infty) \rightarrow [0, \infty)$ is continuous, nondecreasing with:

- $\psi(0) = 0$
- $\psi(t) > 0$ for all $t > 0$
- $\lim_{t \rightarrow \infty} \psi(t) = \infty$

Examples of ψ : $\psi(t) = at$, $\psi(t) = \ln(1 + t)$, $\psi(t) = \frac{t}{1+t}$.

Generalizes Banach contraction: allows slower decay than linear.

Types of Weakly Contractive Mappings

Definition (Weakly $(\varphi - \psi)$ -Contractive)

Type (I): For all $x, y \in X$,

$$d(fx, fy) \leq \varphi(d(x, y)) - \psi(d(x, y))$$

where φ, ψ are continuous nondecreasing with:

- (C1) $\varphi(0) = \psi(0) = 0$
- (C2) φ, ψ positive on $(0, \infty)$
- (C3) $\varphi(t) - \psi(t) < t$ for all $t > 0$

Type (II): Same inequality with conditions (D1)-(D3) where ψ is nonincreasing.

Key advantage: Existence without uniqueness assumption (some theorems).

Theorem 5.169-5.171: Weakly Contractive Fixed Points

Theorem (Harjani-Sadarangani [267])

If (X, d) is complete metric space and $f : X \rightarrow X$ is continuous, nondecreasing with

$$d(fx, fy) \leq d(x, y) - \psi(d(x, y)) \quad \text{for } x \geq y$$

where ψ satisfies (C1)-(C3), and exists $x_0 \in X$ with $x_0 \preceq f(x_0)$, then f has a fixed point.

Theorem (Type (II) Version)

Same conclusion holds for conditions (D1)-(D3) with ψ nonincreasing.

Proof strategy: Show $\{\rho_n\}$ sequence of distances is decreasing and converges to 0.

Theorem 5.172-5.175: Uniqueness Conditions

Theorem (Uniqueness in Ordered Sets)

Under hypotheses of Theorem 5.171 with conditions:

- ① (SC1) *Every pair has lower or upper bound*
- ② (SC2) *Sequential comparability property*
- ③ (SC3) *f maps comparable elements to comparable elements*

Then fixed point is unique.

[Non-uniqueness] $X = \{(1, 0), (0, 1)\} \subset \mathbb{R}^2$ with $\varphi(t) = \frac{2}{3}t + \frac{1}{3}t^2$, $\psi(t) = \frac{2}{3}t$ satisfies weak contraction conditions with two fixed points in X .

Example 5.51: Chain Analysis

Let $X = \mathbb{R}$ with usual order and $\mathcal{C} = \{0\} \cup \{\pm 2^{-n} : n \in \mathbb{N}\}$.

Define $f(x) = \frac{1}{2}x$ for all $x \in \mathcal{C}$.

Then $\mathcal{C}$ is a chain in X and (X, d) complete metric space. The mapping f is:

- Nondecreasing on comparable elements
- Weakly $(\varphi - \psi)$ -contractive with $\varphi(t) = \frac{2}{3}t + \frac{1}{3}t^2$, $\psi(t) = \frac{1}{3}t$
- Unique fixed point at $0 \in \mathcal{C}$

Theorem 5.155-5.156: Multiple Fixed Point Methods

Theorem (Three Distinct Fixed Points)

If there exist points $\underline{y}_1 < \underline{y}_2 < \bar{y}_1 < \bar{y}_2$ such that:

- $\underline{y}_1 \leq F(\underline{y}_1), F(\underline{y}_2) \leq \underline{y}_2$
- $\bar{y}_1 \leq F(\bar{y}_1), F(\bar{y}_2) \leq \bar{y}_2$
- F continuous, strongly increasing with relatively compact range

then F has at least three distinct fixed points x, x_1, x_2 with $\underline{y}_1 \leq x_1 \leq \underline{y}_2, \bar{y}_1 \leq x_2 \leq \bar{y}_2$.

Corollary 5.34: Under same conditions, $x_1 \ll x_2 \ll x_3$ (three ordered fixed points).

Theorem 5.159-5.161: Fixed Points in Function Spaces

Theorem (Function Space Fixed Points)

Let $F : (0, c)_\infty \rightarrow (0, c)$ be continuous with relatively compact range. If there exist a, b, d with $0 < a < b \leq c$ and closed $\Omega_1 \subseteq \Omega$ such that:

- $|Fx(t) - Fx(s)| \leq b - a$ for $x \in (0, c)_\infty$ and $t, x \in \Omega_1$
- $Fx(t) > a$ and $F(x(t)) > a$ when $a \leq x(s) \leq b$ and $s \in \Omega_1$

then $Fx = x$ with $x(t) > a$ for each $t \in \Omega_1$ has a solution.

Application: Boundary value problems and periodic solutions.

Theorem 5.162-5.163: Annular Region Theorems

Theorem (Sandwich Method - Gustafson-Schmitt [260])

] Let $D = \{x \in K : r \leq \|x\| \leq R\}$ (annular region).

If $F : D \rightarrow K$ is continuous mapping bounded sets to precompact sets with:

① $\|x\| = R, Fx = \eta x \Rightarrow \eta \leq 1$

② $\|x\| = r, Fx = \eta x \Rightarrow \eta \geq 1$

③ $\inf_{\|x\|=r} \|Fx\| > 0$

then F has fixed point in D .

Theorem (Reversed Boundaries)

Same conclusion when (i) and (ii) inequalities reversed.

Numerical Example: Fixed Point Iteration

Consider $Tx = \sin(x) + 0.3x$ on $[0, 2]$ with $K = [0, \infty)$.

```
# Fixed point iteration
import numpy as np
import matplotlib.pyplot as plt

def T(x):
    return np.sin(x) + 0.3 * x

# Iteration
x = np.array([0.5])
for i in range(20):
    x = np.append(x, T(x[-1]))

# Check convergence
print(f"Fixed point approximation: {x[-1]:.6f}")
print(f"Verification T(x*) - x*: {T(x[-1]) - x[-1]:.6f}")
print(f"Error: {abs(x[-1] - T(x[-1])):.2e}")

# Convergence rate
rates = np.abs(np.diff(x))
print(f"Convergence rate: {rates[-1]/rates[-2]:.4f}")
```

Theorem 5.164-5.166: Asymptotic Derivative Methods

Theorem (Spectral Radius Bounds)

If $F : K \rightarrow K$ maps bounded sets to precompact sets, satisfies:

- ① $Fx = \eta x \Rightarrow \eta \leq 1$ for $\|x\| = R$
- ② $Fz = \eta Fz \Rightarrow \eta < \infty$ for some $z \in K, \|z\| = R$
- ③ Continuous asymptotically linear majorant with spectral radius < 1

then F has fixed point $u \in K$ with $r \leq \|u\| \leq R$.

Theorem (Fréchet Derivative Condition)

If $F'(0)$ has eigenvector $h \in K$ with $\eta_0 > 1$ and $F'(\infty)$ has spectral radius < 1 , then nonzero fixed point exists.

Theorem 5.167: Agarwal-Regan Multivalued Result

Theorem (Agarwal-Regan [6])

] Let E be Banach space, $C \subseteq E$ cone with $\mathcal{K}(C)$ family of nonempty compact subsets of C .

If $A : \Omega_r \cap C \rightarrow \mathcal{K}(C)$ is upper semicontinuous compact map with one of conditions:

- ① (A) $\|y\| \leq \|x\|$ for all $y \in A(x)$ and $x \in \partial\Omega_R \cap C$, and $\|y\| > \|x\|$ for $x \in \partial\Omega_r \cap C$*
- ② (B) Inequalities reversed*

then A has fixed point in $C \cap (\bar{\Omega}_R \setminus \Omega_r)$.

Extends to multivalued operators and set-valued mappings.

Definition 5.92-5.93: Contractions in Banach Algebra

Definition (Nonlinear Contraction)

A mapping T on Banach space X is nonlinear contraction if:

$$\|Tx - Ty\| \leq \varphi(\|x - y\|)$$

where $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ continuous with $\varphi(r) < r$ for $r > 0$.

Definition (D-Lipschitzian)

A mapping T is D-Lipschitzian if there exists continuous nondecreasing $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $\varphi(0) = 0$ such that:

$$\|Tx - Ty\| \leq \varphi(\|x - y\|) \quad \text{for all } x, y \in X$$

Every Lipschitzian is D-Lipschitzian; converse may not hold.

Theorem 5.176-5.177: Krasnoselski in Banach Algebra

Theorem (Krasnoselski [346])

- *Famous Result*] Let S be closed, convex, bounded subset of Banach algebra X .

If $A, B : S \rightarrow X$ are operators with:

- 1 A is contraction
- 2 B completely continuous
- 3 $Ax + By \in S$ for all $x, y \in S$

then $Ax + Bx = x$ has solution in S .

Theorem (Burton [126])

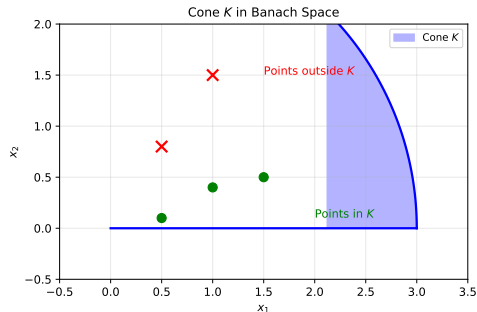
- *Modification*] Condition (c) can be replaced by weaker condition: A contraction and $B : S \rightarrow X$ completely continuous.

Application: Nonlinear integral equations.

Key Theorems Summary Table

Theorem	Space Type	Operator Type	Result
5.145 (Krasnoselski)	Ordered Banach	Positive Linear	Characteristic Vector
5.150 (Aman)	Ordered Banach	Isotone	Minimal Fixed Point
5.155	Ordered Banach	Increasing	Three Fixed Points
5.162	Cone in Banach	Continuous	Annular Region
5.168 (Dhage)	P. Ordered	Hybrid	Solution of $\mathcal{P}x + \mathcal{Q}x = x$
5.171-5.175	P. Ordered	Weakly $(\varphi - \psi)$	Unique Fixed Point
5.176 (Krasnoselski)	Banach Algebra	A contr., B c.c.	Solution in S

Cone Properties and Fixed Point Existence



Positive Operator Properties

1. Positive: $T(K) \subseteq K$ (maps cone to itself)
2. Strictly Positive: $T(\bar{K}) \subset K$ (interior)
3. Strongly Positive: $T(\bar{K}) \subseteq \text{int}(K)$

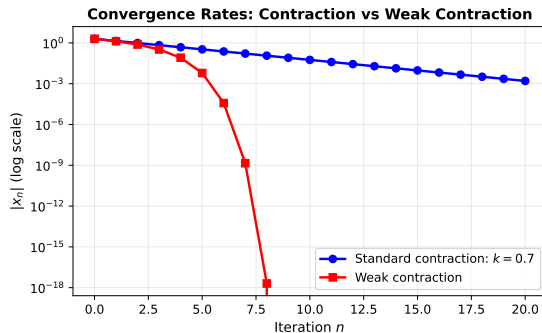
Fixed Point Properties:

- Every positive linear operator has a characteristic vector
- Fixed points exist under compactness conditions
- Multiple fixed points possible (order-theoretic methods)

Geometric interpretation:

- Positive operator maps cone into itself
- Fixed points lie in cone (order-theoretic nature)
- Compactness ensures convergent subsequences

Convergence Analysis



Weakly Contractive Mapping

Definition: For every x, y in X ,

$$\|Tx - Ty\| \leq \|x - y\| - \psi(\|x - y\|)$$

where $\psi: [0, \infty) \rightarrow [0, \infty)$ satisfies:

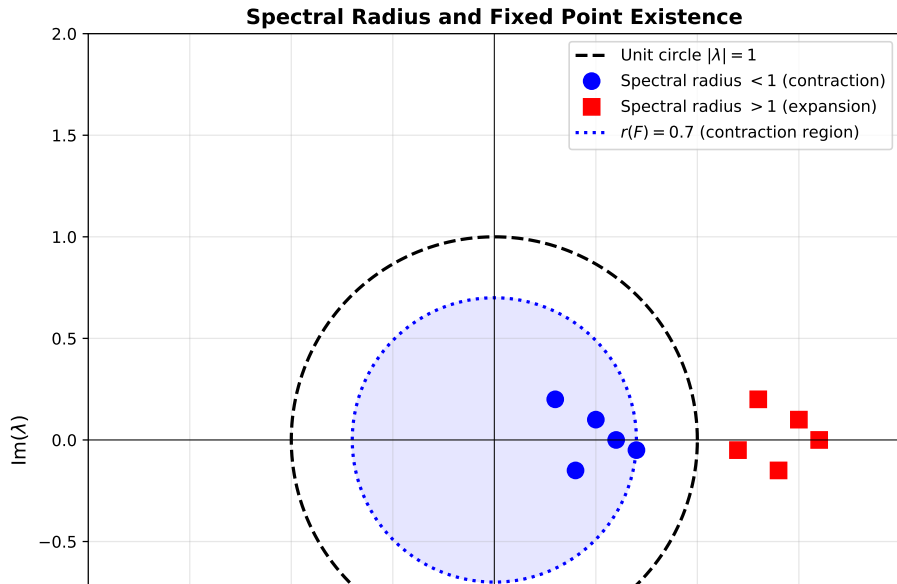
- $\psi(0) = 0$
- ψ continuous and nondecreasing
- $\psi(t) > 0$ for all $t > 0$
- $\lim_{t \rightarrow \infty} \psi(t) = \infty$

Examples: $\psi(t) = at$, $\psi(t) = \ln(1+t)$

Weak vs. standard contractions:

- Weak contraction: $\|Tx - Ty\| \leq \|x - y\| - \psi(\|x - y\|)$
- Allows slower convergence rates
- Useful for applications with nonlinear contractions

Spectral Radius and Stability



Partially Ordered Sets: Concepts and Structure

Fixed Point Theorems in Partially Ordered Sets

Key Concepts:

1. Partially Ordered Set $(X, \leq)$:
 - Reflexive: $x \leq x$
 - Transitive: $x \leq y, y \leq z \Rightarrow x \leq z$
 - Anti-symmetric: $x \leq y, y \leq x \Rightarrow x = y$
2. Monotone (Isotone) Mapping $f: X \rightarrow X$
 - Nondecreasing: $x \leq y \Rightarrow f(x) \leq f(y)$
 - Nonincreasing: $x \leq y \Rightarrow f(x) \geq f(y)$
3. Weakly Contractive Mapping:
 $\|Tx - Ty\| \leq \|x - y\| - \psi(\|x - y\|)$
where $\psi(0) = 0$, ψ continuous, nondecreasing
4. Convergence Theorem:
If $x_0 \leq f(x_0)$, then $x_n = f(x_{n-1})$ is nondecreasing
and converges to fixed point x^*
5. Uniqueness Conditions (SC1)-(SC3):
 - (SC1) Every pair has a lower bound or upper bound
 - (SC2) Sequences have comparable subsequences
 - (SC3) f maps comparable elements to comparable elements

Applications to Boundary Value Problems

Typical application: Periodic boundary value problem

$$x''(t) + f(t, x(t)) = 0, \quad t \in [0, T] \quad (1)$$

$$x(0) = x(T), \quad x'(0) = x'(T) \quad (2)$$

Fixed point formulation:

- Green's function $G(t, s) \geq 0$ defines positive operator
- Integral operator: $Tx(t) = \int_0^T G(t, s)f(s, x(s))ds$
- Theorems 5.150, 5.162, 5.168 guarantee solution existence
- Order-theoretic methods provide constructive iteration

Advantage: Weak differentiability assumptions; handles nonlinearities.

Applications to Integral Equations

Hammerstein integral equation:

$$x(t) = g(t) + \lambda \int_a^b K(t, s)f(s, x(s))ds$$

Operator framework:

- Write as $Fx = x$ with $Fx = Ax + Bx$
- A contraction, B compact in Krasnoselski method
- Application of Theorem 5.176
- For weakly contractive case: Theorems 5.171-5.175

Key advantage: Handles both singular and regular kernels via operator splitting.

Convergence Theorems for Computation

Iteration method: $x_{n+1} = Tx_n$

Convergence guarantees:

- 1 **Banach:** Linear convergence if $\|T'(x^*)\| < 1$
- 2 **Monotone:** Guaranteed if T isotone and $x_0 \leq Tx_0$
- 3 **Weak contraction:** Guaranteed via distance decay $\rho_n \leq \varphi(\rho_{n-1}) - \psi(\rho_{n-1})$
- 4 **Hybrid (Dhage):** Combination of linear and nonlinear components

Advantage of order-theoretic methods:

- Provide bounds: $\underline{x} \leq x_n \leq x^* \leq \bar{x}_n$
- Guide computational algorithms
- Useful for error estimation

Chapter 5f: Main Takeaways

Three main perspectives on fixed points:

1 Order-Theoretic:

- Isotone/monotone operators
- Minimal/maximal fixed points
- Amann, Krasnoselski results

2 Metric-Based:

- Contraction principles
- Weakly contractive mappings
- Convergence rates

3 Algebraic:

- Banach algebra structure
- Krasnoselski hybrid method
- Operator splitting techniques

Unified framework: All methods apply to practical problems via appropriate operator construction.

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Fixed Point Theorems in Ordered Banach Spaces

Applications to approximation of general mappings

Complete theoretical framework for modern nonlinear analysis